

Master's Project Report
PH-595
**Constraining Equation of State of Neutron
Stars using Neutron Star-Black Hole
mergers**

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May 04, 2025

Acknowledgements

I would like to express my sincere gratitude to my guide, Dr. Rahul Kashyap, for his invaluable guidance, insightful suggestions, and continuous encouragement throughout the course of this project. His support has been instrumental in shaping this work.

I am also deeply thankful to my peer group members, Mr. Yugesh Bhoge for his consistent support, cooperation, and constructive discussions that enriched my understanding and efforts.

I extend my heartfelt thanks to my friends Gourab, Tushar, Saikat, and Satyajit, whose motivation and companionship provided great moral support throughout this journey.

A special note of appreciation goes to the Spacetime Computer Cluster at IIT Bombay and CalTech LIGO Cluster, without which the computational work involved in this project would not have been possible.

Lastly, I am profoundly grateful to my family for their unwavering support, love, and encouragement, which have been my strength throughout this endeavour.

Abstract

Neutron stars and their internal structure remain one of the most intriguing and fundamental problems in modern astrophysics. This project aims to constrain the equation of state (EoS) of neutron stars by utilizing gravitational wave (GW) signals generated during compact object mergers. The primary focus is on Neutron Star-Black Hole (NSBH) mergers, which offer a cleaner environment for analysis compared to Binary Neutron Star (BNS) mergers.

A key objective of this study is to understand how the tidal deformability of neutron stars depends on their underlying EoS, and to leverage GW observations to infer properties of dense matter physics. To optimize computational efficiency, Monte Carlo parameter studies were conducted to balance the trade-off between computational cost and the validity of the results.

Additionally, different waveform approximants were explored, including `IMRPhenomTidalv2` for symmetric mass systems with free tidal deformability parameters, and `IMRPhenomNSBH` for general NSBH analyses. The performance of these models was assessed based on their ability to recover physical parameters from simulated merger events.

Finally, variations in neutron star EoS, specifically `APR4`, `SLy`, and `DD2`, were investigated to study their impact on gravitational wave signals. Posterior distributions and relative errors were compared across different EoS models and across multiple detectors to assess the sensitivity and robustness of parameter recovery.

Current Status of the Research

In the previous semester, we began by studying the fundamental theory of merger events, computational methods, and preliminary data analysis. Our primary objective is to investigate the **neutron star equation of state (EoS)** using gravitational wave (GW) data from neutron star–black hole (NSBH) mergers. This study focuses on the **tidal deformability** of neutron stars, which is directly influenced by the choice of EoS.

As the project progressed, we employed **Bayesian inference** techniques and utilized the **Bilby** framework for parameter estimation and model selection. The Bilby pipeline requires the selection of appropriate waveform models or approximants. For nearly symmetric binary systems, we used `IMRPhenomNRTidalv2`, while for general asymmetric NSBH systems, `IMRPhenomNSBH` proved to be a more suitable waveform approximant.

Using LIGO data, we compared different EoS models — `SLy`, `APR4`, and `DD2` — across various detector sensitivities (`A#`, `ECC`, and `O5`). Our analysis showed that the `ECC` detector exhibited the best performance among the three configurations. Furthermore, `IMRPhenomNSBH` yielded more accurate results compared to `IMRPhenomTidalv2`.

In the current semester, our goal is to obtain more precise results while minimizing computational costs. To achieve this, we aim to optimize the Monte Carlo parameters for improved efficiency. For highly asymmetric systems, we will use `IMRPhenomNSBH` with the black hole tidal deformability fixed at $\Lambda_{\text{BH}} = 0$ [1]. For low-mass or nearly symmetric binaries ($0.6 < q < 1.0$), we will allow both tidal deformabilities to vary freely.

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Chapter 1

Introduction

Till now, the understanding of the structure of a neutron star is not known precisely. Different equations of state try to model the neutron stars. The main goal of this work is to differentiate the signals of Binary Neutron Star (BNS) mergers and Neutron Star Black Hole (NSBH) mergers with the assistance of constraining equation of state of neutron stars. As we know that till the date, we have only 2 observations of BNS mergers and NSBH mergers each, which are very few in comparison to Binary Black Hole (BBH) mergers. So the properties of these systems, especially of the neutron stars, are very unclear. The BNS mergers emit both gravitational waves as well as their electromagnetic counterparts (though not all detectable), the study of tidal structure of neutron stars from these systems is difficult. We can get this knowledge from the NSBH mergers as the tidal deformability of a black hole is zero.

Matur et al. 12 compared different spinning and non-spinning models and found that the spinning model signatures are merging about $2.5ms$ earlier to the non-spinning model. In addition to that, the amplitude for softer EoS tend to be higher in both, after- and pre-merger, phases. The TM1a0 and SFHxa0 models corresponding to stiffest and softest EoS, respectively, are claimed to constrain the EoS for high SNR.

Binary systems are the systems of two stars that are bounded gravitationally. They can be Binary Neutron Stars (BNS), Binary Black Holes (BBH), Neutron Star-White Dwarf (NSWD) or Neutron Star-Black Hole (NSBH), or even two main sequence stars also. Their evolution is categorised in 3 main phases: **Inspiral**, in which both the bodies bounded closely and revolve around the common centre of mass; **Merger**, in which the two bodies collide and leave the compact remnant; and **Ringdown**, where the remnant body tries to stabilize itself by emitting gravitational radiations.

One of the main issues with ground-based telescopes is to discover the coalescence of BH-NS binaries accompanied by the tidal disruption and followed by the electromagnetic spectrum. If the neutron star is not disrupted, the neutron star will behave as a point particle and the system will be indistinguishable with the highly asymmetric binary black hole system[2].

The gravitational wave signals depend on the masses of the binary components. But the NS and BH can't have any random mass. The stellar evolution shows that a non-spinning neutron star can have a maximum mass of $2.7^{+0.4}_{-0.5}M_{\odot}$ and minimum black hole mass can't go below $5.4^{+1.0}_{-0.7}M_{\odot}$ [15]. These findings clearly show [a mass gap between the masses of NS and BH](#). With next-generation detectors and hence more detections, the accuracy can be improved.

Previous studies have shown that for lower mass ratios ($q = m_2/m_1$, where m_1 and m_2

are masses of black hole and neutron star, respectively), less mass is ejected and system behaves like a BBH system. The final fate of the system depends on the mass ratio and the compactness of the neutron star. The lower mass ratios and lower compactness, hence higher tidal deformability, results in large Neutron Star (NS) disruption. The effects on tidal disruption depend on the EoS used. Using the polytropic equation of state (EoS), i.e., $P = \kappa \rho^\Gamma$, the [stiffer EoS shows a more massive, larger and longer-lived tidal tail](#). For Stiffer EoS, the NSs have larger radius which imply less compactness, hence, stronger disruption. One important note to take here is that NSBH with low compactness are the best candidates to investigate the signatures from NSBH mergers [\[14\]](#).

In different binary systems, the tidal disruption of neutron stars happens differently. For high mass ratios, there is more probability of disruption in comparison to low mass ratios. The spins of black hole in the binary system can also be the reason of different kind of tidal disruption events. If the black hole is **progradly** rotating (aligned rotation with respect to the orbital motion), the higher spin magnitudes enhances the NS disruption and leads to more remnant mass, in comparison to **retrograde** motion.

One of the main aim of this work is to distinguish the mixed and neutron stars binaries. [Chen and Chatziioannou 4](#) introduced a novel technique to identify NSBH in a population of low-mass compact binary mergers using only gravitational wave data. They used a hierarchical mixture model to measure the ratio of NSBHs to BNSs in a population. If the NSBH is misidentified as a BNS, it would appear as an outlier in the NS properties measurements.

A different study proposes a novel data analysis strategy that combines Bayesian inference with machine learning techniques to differentiate between BNS and NSBH mergers using GW signals. Tidal deformability serves as a key diagnostic parameter—since black holes have effectively zero tidal deformability, NSBH systems can in principle be distinguished from BNS systems. However, challenges arise when black hole masses closely resemble those of neutron stars, resulting in degenerate GW signatures. Accurate classification of such systems is crucial for understanding the astrophysical population of compact binaries and constraining the neutron star equation of state (EoS). With enhanced detector sensitivity and increasing event numbers, the proposed methodology has strong potential to improve classification and parameter recovery in future GW observations[\[6\]](#).

Before diving into the basic theoretical part of this project, let's first answer some questions that are likely to come up later:

- What do we mean by constraining an equation of state?

[It means to infer the allowed physical properties like mass, radius, tidal deformability and compactness etc. By comparing the theoretical models with the observational gravitational waves signals, we can narrow down the possible functional form of the EoS. Different EoS will predict different \$\Lambda\$ -values. By accurately measuring them, we can rule out some EoS and favor others.](#)

- What is the purpose of this project?

[We are using simulated gravitational waves data from NSBH mergers to extract \$\Lambda\$ of NS. NSBH mergers allow cleaner measurements of \$\Lambda\$ in comparison to BNS mergers. So, our goal is to recover the injected parameters precisely using the Bayesian Parameter Inference \(BILBY\).](#)

Let us now delve into the detailed theory of the EoS and Tidal Deformability.

Chapter 2

GW Inference of the Λ_{NS}

2.1 Longterm evolution of binary system[10]

The inspiral phase of the binary system can be considered as a stepwise advancement towards the merger phase. The changes in the structure of the objects at every step are very small so that we can take the system at those steps to be in equilibrium. Although the orbit is constantly shrinking due to gravitational wave emission, the evolution of stars is nearly constant in comparison to the dynamical scale of the orbit. These sequences provide a precise study of the tidal effects on neutron star just before its merging into the black hole.

- the eccentricity of the orbit decreases very rapidly ($e \propto a^{-19/12}$) to an asymptotically circular orbit. So, just before merger, we can take the orbit of BH-NS merger to be circular.
- the neutron star is unlikely to be tidally locked, because the effects of viscosity are likely to be insufficient. Thus, the spin of the neutron star can affect the merger dynamics significantly only if the rotational period is extremely short at the outset and the spin-down is not severe.

2.2 Tidal Deformability

Tidal Deformability Λ , as name suggests, is defined as the quantity that corresponds to the deformation of a body due to some tidal field. It is a consequence of the gradient of gravitational force. It quantifies the non-equilibrium between the self gravity of the stellar object and tidal force due to other. It measures the object's ability to redistribute its mass and deform under the influence of these forces. Mathematically, it can be defined as

$$\Lambda = \frac{k_2}{C^5} \quad (2.1)$$

where k_2 is known as the **Love Number**¹, and C is the **Compactness** of the star and is defined in terms of mass (M) and radius (R) of the star as:

¹For planets and stars, these numbers measure the resistance of their material against the tidal forces, and the resulting contribution to their gravitational multipole moments.

$$C = \frac{GM}{Rc^2} \quad (2.2)$$

According to general relativity, nonrotating deformed black holes, instead, show no addition to their gravitational multipole moments, and all of their Love numbers are zero[5].

Neutron Stars, Black Holes and White Dwarfs are highly compact stars than the main sequence stars like our Sun. Neutron Star experiences finite tidal deformation because of non-trivial internal structure. It depends on the EoS, the mass and the radius. It is a property assigned to matters, but black holes have no internal structure. The mass of BH represents the curvature of spacetime, entirely described by the geometry of Spacetime around the BH. The event horizon of a black hole behaves in a way that it remains spherically symmetric under external tidal forces and hence its tidal deformability is taken to be zero ($\Lambda_{BH} = 0$).

Effective Tidal Deformability

The GW signals of the binary systems gives all the information about the system we can get. The late-inspiral phase of the system has the prints of tidal interactions between the binary components. These effects can be primarily quantified as **Effective Tidal Deformability**, $\tilde{\Lambda}$. This is a specific combination of individual tidal deformabilities Λ_1 and Λ_2 and given as²:

$$\tilde{\Lambda} = \frac{16}{13} \frac{(m_1 + 12m_2)m_1^4\Lambda_1 + (m_2 + 12m_1)m_2^4\Lambda_2}{(m_1 + m_2)^5} \quad (2.3)$$

For NSBH system, $\Lambda_1 = \Lambda_{BH} = 0$ and $\Lambda_2 = \Lambda_{NS} \neq 0$. So, we can write effective tidal deformability of the system as:

$$\tilde{\Lambda} = \frac{16}{13} \frac{(m_2 + 12m_1)m_2^4\Lambda_2}{(m_1 + m_2)^5}$$

$$\tilde{\Lambda} = \frac{16}{13} \frac{(q + 12)q^4}{(1 + q)^5} \Lambda_2 \Rightarrow \Lambda_2 = \frac{13}{16} \frac{(1 + q)^5}{(q + 12)q^4} \tilde{\Lambda} \quad (2.4)$$

Then, we can find the tidal deformability of the Neutron Star from the GW signal as shown above.

Now, as we can see, the mass ratio q can be set to same value for different masses. For example, $(3.0, 1.5)M_\odot$ and $(5.0, 2.5)M_\odot$ will have same q -value ($=0.5$). So, with this point taking into account, we can define another mass quantity called **Chirp Mass**, $\mathcal{M}$:

$$\mathcal{M} = M\eta^{3/5} = \frac{(m_1 m_2)^{3/5}}{(m_1 + m_2)^{1/5}} \quad (2.5)$$

We can find the orbital frequency f as well as the rate of change of orbital frequency $\dot{f}$ from which we can calculate the chirp mass[11]:

²As defined earlier, m_1 and Λ_1 are the properties of primary component, while m_2 and Λ_2 are that of secondary component. Also $m_1 > m_2$

$$\mathcal{M} = \frac{c^3}{G} \left(\frac{5}{96} \pi^{-8/3} f^{-11/3} \dot{f} \right)^{5/3} \quad (2.6)$$

We will discuss more about the GW parameters in Sec-3.2

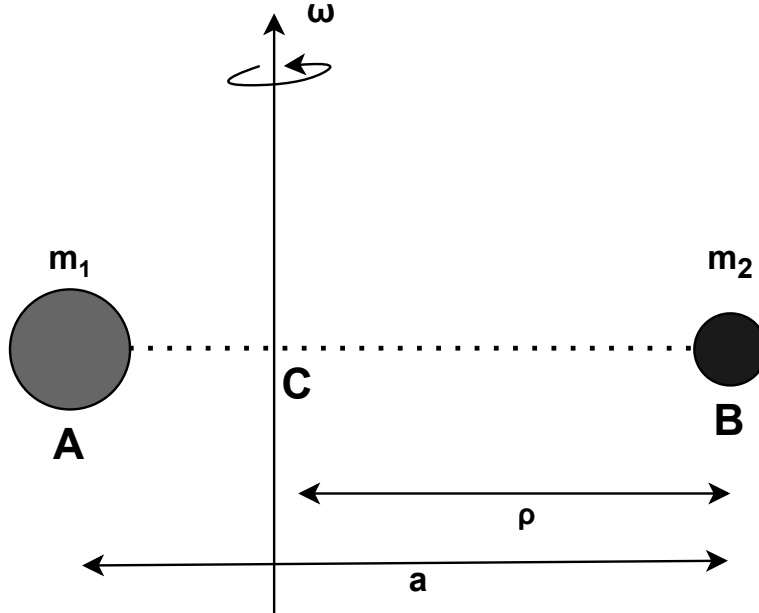
2.3 Mass-Shedding Limit[10]

To deeply understand the tidal deformation, let's discuss about the mass-shedding limit of a star. It defines the tidal deformation condition of a star, specifically neutron star. It depends on the mass ratio of the system (the binary stars that are orbiting around each other).

Taking NSBH system as an example, the mass shedding from the NS occurs when the self gravity of the NS is overcome by the tidal force exerted by the black hole. The condition is approximately given by:

$$\frac{2GM_{BH}(c_R R_{NS})}{r^3} \sim \frac{GM_{NS}}{(c_R R_{NS})^2} \quad (2.7)$$

here, c_R is known as the degree of tidal elongation of the stellar radius. To understand it better, let's take an example of two stars A and B, with masses m_1 and m_2 , respectively, revolving around their center of mass (CoM) C with angular frequency ω , as shown below:



With simple calculation, we can write:

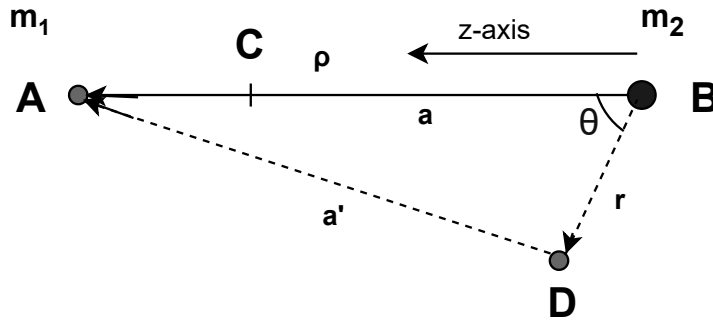
$$\rho = \frac{m_1}{M} a \quad (2.8)$$

where $M = m_1 + m_2$. In the CoM frame of reference, both stars will be stationary. Neither they will come towards, nor go away from the CoM. So the net (gravitational + centrifugal) force will be zero.

$$F_{gravitational} = \frac{Gm_1m_2}{a^2} = m_2\omega^2\rho = F_{centrifugal}$$

$$\omega^2 = \frac{Gm_1}{a^2\rho} \quad (2.9)$$

Now, instead of considering the stars as point object, let's assume there are mass distributions for both the stars with their respective center of masses at A and B. Considering D be any point in m_2 mass distribution.



The centrifugal acceleration (along the z-axis) can be written in terms of *grad* of a potential as follows:

$$\mathbf{g}_c = -\omega^2\rho\hat{\mathbf{z}} = -\nabla\chi' \quad (2.10)$$

As $z = r\cos\theta$ and $\cos\theta$ can be written as $P_1(\cos\theta)$. Using 2.9:

$$\chi' = \frac{Gm_1}{a} \frac{r}{a} P_1(\cos\theta) \quad (2.11)$$

Also, gravitational acceleration at point D due to m_1 :

$$\mathbf{g}_g = -\nabla\phi' \implies \phi' = -\frac{Gm_1}{a'} \quad (2.12)$$

With a bit of algebra, it can be approximated as:

$$\phi' \approx -\frac{Gm_1}{a} \left[1 + \left(\frac{r}{a}\right) P_1(\cos\theta) + \left(\frac{r}{a}\right)^2 P_2(\cos\theta) \right] \quad (2.13)$$

So, net potential due to all external force acting on the mass element D will be:

$$\chi = \chi' + \phi' = -\frac{Gm_1}{a} \left[1 + \left(\frac{r}{a}\right)^2 P_2(\cos\theta) \right] \quad (2.14)$$

From the above expression, we can infer that χ is constant in first order in (r/a) . The first order variation cancels each other. Away from the center, $\mathbf{g}_c$ remains constant but this is not same for $\mathbf{g}_g$. $\mathbf{g}_c < \mathbf{g}_g$ towards A and vice-versa. Hence the mass distribution of B will be the subject of residual acceleration which is the reason of **Tidal Elongation**.

After understanding the tidal elongation, we can define the mass-shedding limit as the orbit at which Eq-2.7 is approximately saturated:

$$r_{ms} := 2^{1/3} c_R \left(\frac{M_{BH}}{M_{NS}} \right)^{1/3} R_{NS} \quad (2.15)$$

Now, as they orbit around each other, the orbital separation will decrease due to the gravitational wave radiations. The **Tidal Disruption Event** (TDE) occurs at $r < r_{ms}$. The electromagnetic counterparts of TDE generated only if it occurs outside the innermost stable circular orbit (r_{ISCO}) of the black hole. Otherwise, there will be no remnant disk or ejecta formed. So,

$$r_{ms} > r_{ISCO} \quad (2.16)$$

This r_{ISCO} is a sensitively decreasing function of the dimensionless spin χ and defined as:

$$r_{ISCO} = \hat{r}_{ISCO}(\chi) \frac{GM_{BH}}{c^2} \quad (2.17)$$

and hence,

$$\frac{r_{ms}}{r_{ISCO}} = \frac{2^{1/3} c_R}{\hat{r}_{ISCO}(\chi)} \left(\frac{M_{BH}}{M_{NS}} \right)^{-2/3} \left(\frac{GM_{NS}}{c^2 R_{NS}} \right)^{-1} \quad (2.18)$$

From here, we can say that for tidal disruption to occur, the system will have to follow Eq-2.16 which can be achieved by:

- low mass ratio, $q = \frac{M_{BH}}{M_{NS}}$
- low compactness, $\mathcal{C} = \left(\frac{GM_{NS}}{c^2 R_{NS}} \right)$
- high spin χ

2.4 Neutron Star Equation of State

The equation of state of Neutron Star defines a relation between pressure P and energy density ϵ , or equivalently, mass density ρ inside the star. It encodes the response of dense nuclear matter under extreme gravitational and degeneracy pressure and plays a crucial role in determining observable neutron star properties such as:

- $M - R$ relation,
- Compactness, $\mathcal{C} = \frac{GM}{Rc^2}$,
- Tidal Deformability, $\Lambda = \frac{2}{3} \frac{k_2}{C^5}$

In general theory of relativity, an EoS is used as an input to the **Tolman Oppenheimer Volkoff (TOV)** equations, which describe the hydrostatic equilibrium of a spherically symmetric star. Because direct probes of the NS interior are impossible, astrophysical observations — especially through gravitational waves — provide indirect but powerful constraints on the EoS.

Equation of States are classified in two categories:

1. **Softer EoS:** It results in more compressible NS, with smaller radius and lower Tidal Deformability.
2. **Stiffer EoS:** Due to stiffness of Eos, it is less compressible and hence have larger radius with higher Tidal Deformability and higher maximum masses.

To span a representative range of nuclear matter behavior, we use three realistic and commonly used EoS models:

Model	Type	Maximum Mass	$R_{1.4}$	$\tilde{\Lambda}_{1.4}$
APR4	Soft	$\sim 2.2 M_{\odot}$	~ 11 km	~ 300
SLy	Moderately Soft	$\sim 2.05 M_{\odot}$	~ 11.6 km	~ 350
DD2	Stiff	$\sim 2.42 M_{\odot}$	~ 13 km	~ 700

These models allow us to test how different nuclear matter assumptions affect gravitational-wave observables, particularly the tidal deformability Λ_2 , and to study how well it can be recovered in NSBH systems.

These theoretical insights into tidal deformability, mass-shedding limits, and their dependence on mass ratio and equation of state form the foundation for our analysis. To extract these physical parameters from gravitational wave signals, we now turn to the tools of **Bayesian inference** and numerical **parameter estimation** pipelines. In the next chapter, we outline the methodology adopted for simulating signals, setting priors, generating waveforms, and performing inference using the **BILBY framework**.

Chapter 3

Methodology

3.1 Bayesian Inference

Bayesian inference is a method that for a given evidence, predicts the possibility of a hypothesis. As the name suggests, it uses Bayes' theorem in its root in which without observing the event in concern, we first assume a *prior*, and based on this we try to find the desired probability. In general, Bayes' theorem says,

$$P(H|E) = \frac{P(E|H)P(H)}{P(E)} \quad (3.1)$$

Here, $P(H)$ is prior probability, $P(E|H)$ is the likelihood of the event provided that the assumed hypothesis is correct, and $P(E)$ is known as Evidence. Using them, we keep updating the probability of the hypothesis as more evidence becomes available and make decisions predicted by posteriors.

3.1.1 Prior, Posterior and Likelihood

So, let's see first what are *prior*, *likelihood* and *evidence* and what *hypotheses* we have to use for our purpose. Let's modify Eq-3.1:

$$p(\theta|d) = \frac{L(d|\theta)\pi(\theta)}{Z} \quad (3.2)$$

θ : parameters of the model that we are taking into account,

d : measurement data,

$p(\theta|d)$: probability density of θ for given d .

In astrophysical sense, $L(\theta|d)$ is a description of the measurement. The form of likelihood is what we choose. For Gravitational-wave astronomy (GWA), we usually assume a Gaussian-noise likelihood function,

$$L(d|\theta) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{1}{2} \frac{|d - \mu(\theta)|^2}{\sigma^2}\right) \quad (3.3)$$

Why we choose Gaussian noise here? Because so many random noises combine to show

nearly a Gaussian-like strain noise as combined likelihood is the product of individual likelihood of each bin [13]. Here, $\mu(\theta)$ is a template for gravitational strain waveform ¹.

Prior ($\pi(\theta)$), as name suggests, is something we choose ourselves before starting the measurements. It can be an educated guess, or just a random choice, it depends on the system that we are considering. θ can take so many parameters of the model we are assuming. For instance, in a binary, there are as much as 15 parameters that define the system. They are categorized into 2 classes:

- **intrinsic parameters:** depends on the intrinsic nature of two bodies.
- **extrinsic parameters:** depends on how we see and locate the binary system from Earth.

Intrinsic Parameters (8)	Extrinsic Parameters (7)
Primary mass (M_1)	Luminosity distance (d_L)
Secondary mass (M_2)	time of coalescence (t_c)
Primary spin (χ_1)	Inclination angle (i)
Secondary spin (χ_2)	Right Ascension (Θ)
Primary Tidal Deformability (Λ_1)	Declination (Φ)
Secondary Tidal Deformability (Λ_2)	Polarization angle (Ψ)
Eccentricity (e)	Phase of coalescence (ϕ_c)
Semi major axis (a)	

Table 3.1: Intrinsic and Extrinsic parameters of a binary system

Constraining all 15 parameters simultaneously will be a very heavy task to complete. Usually, we don't do this. We try to look for just few parameters. For BNS, we look for M_1 , M_2 , Λ_1 , Λ_2 , and for NSBH, we look for M_1 , M_2 , χ_1 and Λ_2 ². To find the posterior distribution of a particular parameter, we marginalize over the other remaining parameters and can obtain marginalized posterior:

$$p(\theta_i|d) = \frac{L(d|\theta_i)\pi(\theta_i)}{Z} \quad (3.5)$$

3.1.2 Evidence, Odds and statistical preference

The denominator in Eq-3.5 is known as Evidence and can be expressed as

$$Z \equiv \int d\theta L(d|\theta)\pi(\theta) \quad (3.6)$$

Evidence is of so much importance when we compare one hypothesis with the other one. It selects the model that fits best and statistically show that by how much the model

¹This template can be expressed as

$$\mu(\theta) = F_+(\Theta, \Phi, \Psi)h_+(\theta) + F_\times(\Theta, \Phi, \Psi)h_\times(\theta) \quad (3.4)$$

where $F_{+,\times}$ are antenna response factors and $h_{+,\times}$ are metric perturbations.

²By convention, we take primary object to be more massive. So for NSBH system, M_1 is the mass of black hole and M_2 is the mass of neutron star. Also, as the tidal deformability of black hole is Zero, so instead of taking Λ_1 into account, we take the spin of black hole χ_1 as the 4th parameter.

is preferred over others. Let's say the evidence for model 1 (binary neutron star with spin) and model 2 (binary neutron star without spin) are Z_{spin} and $Z_{no spin}$, respectively, then the Bayes Factor is

$$BF_{no spin}^{spin} = \frac{Z_{spin}}{Z_{no spin}} \quad (3.7)$$

which by convenience, we write as

$$\log BF_{no spin}^{spin} = \log(Z_{spin}) - \log(Z_{no spin}) \quad (3.8)$$

A threshold of $|\log BF| = 7 - 8$ is used as the level of "strong evidence" to infer that model 1 is much better than model 2 [8]. While calculating the evidences, the number of parameters for each model can be different [13].

3.1.3 Samplers

We can make a prediction about the data d for the given parameters θ . This can be done by gravitational waveforms, also known as **approximants**. For example, IMRPhenomD, SEOBNRv4, IMRPhenomPv2, etc. But the problem is other way around. We have the data, and we want to estimate the parameter values. *The inverse problems are in general very computationally expensive to handle. But there is a solution: samplers.*

Sampling is an stochastic algorithm, in which it generate a list of posterior samples θ picked from the posterior distribution. We can easily generate the marginalized probability of any number of set of parameters and can visualize using corner plots.

The sampling algorithms can be split into 2 categories:

- (a) Markov-chain Monte Carlo (MCMC)
- (b) Nested Sampling

In MCMC method, particle undergoes a random walk through a posterior distribution. The position of particle (more preferably called **walker**) for the next iteration depends on the position on the previous iteration. It compare the possibility of that particular position with others, called *acceptance rule* and try to converge the chain ³.

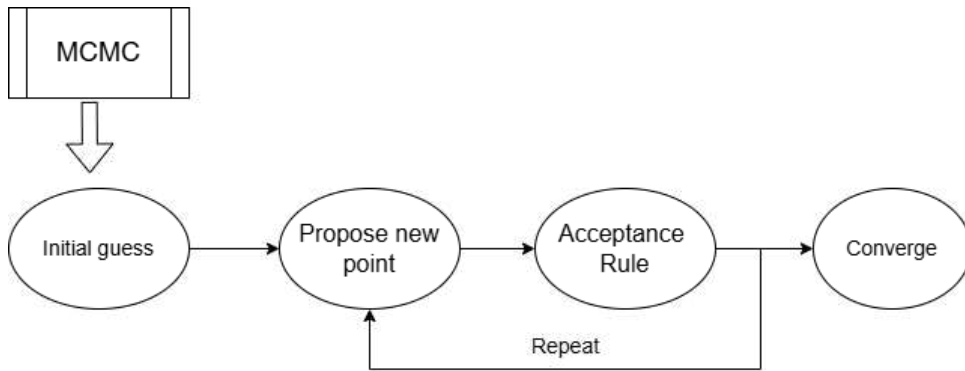


Figure 3.1: MCMC algorithm

For astronomy purposes, **EMCEE** is the best MCMC code.

³It is necessary to have a 'burn-in' phase in the chain. After this phase, the walkers will settle into a steady state and start sampling from the distribution.

Other than MCMC, there is an alternative for this is MCMC, called **Nested Sampling**. Nested sampling simply calculates the evidence values, instead of drawing the samples from the posterior distribution for which MCMC methods are designed.

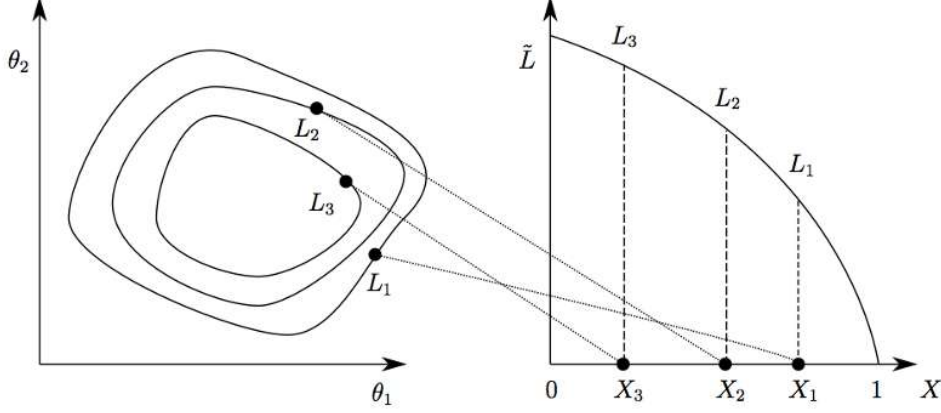


Figure 3.2: Schematic of Nested Sampling
[3]

In the figure above, X_i are prior masses, θ_i are the different parameters. So for different prior masses, there are different contours of likelihood. Each contour is a hyper-surface with constant likelihood. Now let's say, there are 3 prior masses with 3 likelihoods. So, in nested sampling, we first throw away the lowest likelihood point and the corresponding highest prior mass (X_0, L_0) and add a new likelihood surface with a higher likelihood than the removed one. We can possibly estimate an upper limit on the evidence for each iteration, so this can be used to get the termination condition for the algorithm.

3.2 Parameter Estimation using BILBY

Bilby is a user-friendly Bayesian inference library. The aim of bilby is to provide a user-friendly interface to perform parameter estimation. It is primarily designed and built for inference of compact binary coalescence events in interferometric data[1].

The process involves injecting simulated gravitational wave signals into synthetic data, prior definitions, and sampling of posterior distributions samplers.

3.2.1 Injection

The initial step is to inject a simulated gravitational wave signal into synthetic data using predefined parameters⁴:

- Masses: $M_1 \equiv \text{Mass of BH}$, $M_2 \equiv \text{Mass of NS}$
- Spin: $\chi_1 \equiv \text{Spin of BH}$, $\chi_2 \equiv \text{Spin of NS}$ (very small, nearly zero)
- Tidal Deformabilities: $\Lambda_1 (\approx 0)$, Λ_2
- All other parameters like Luminosity distance D_L , RA, Dec, phase and angles corresponding to orientation of the system, will be fixed.

⁴Here we are constraining 4-5 parameters only for initial runs, instead of whole 15-17 parameter space

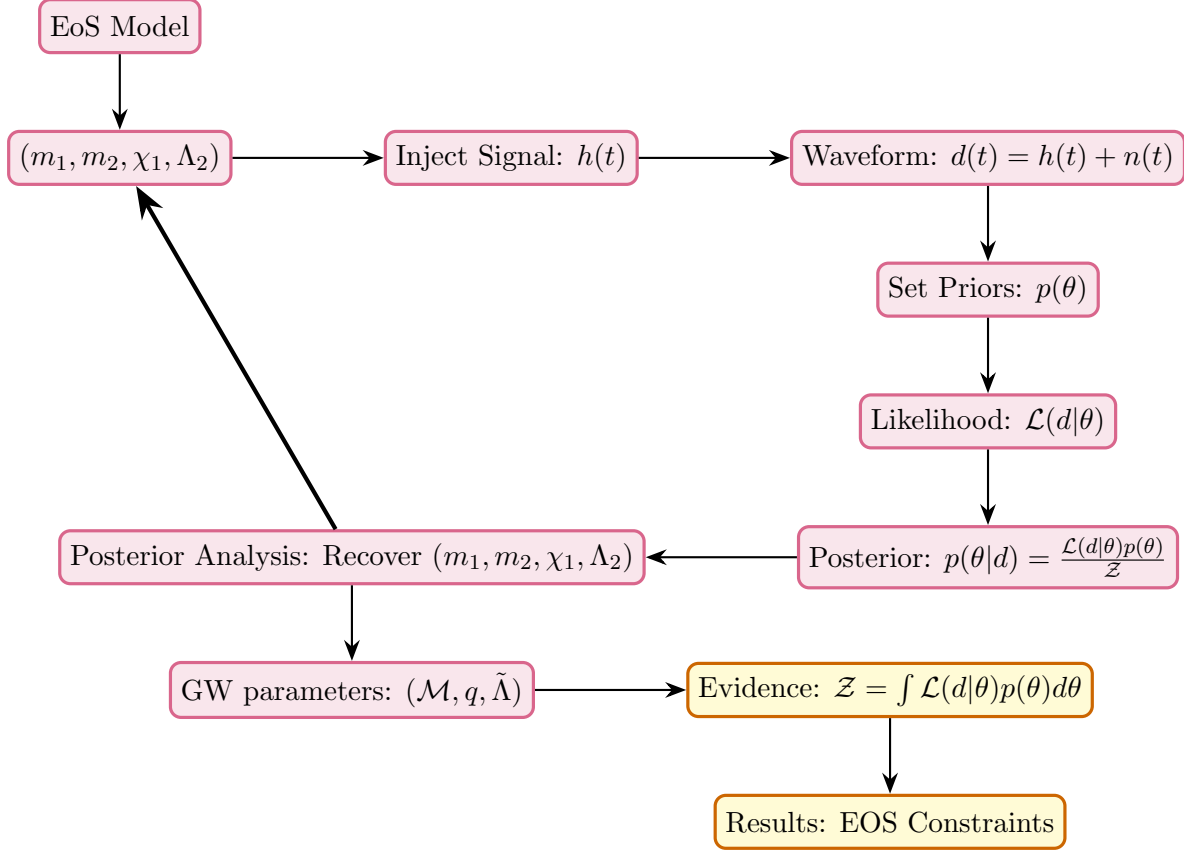


Figure 3.3: Concept Map of Methodology for NSBH EOS Constraints

3.2.2 Waveform Generation

After the injection, the waveform is generated using the waveform approximants. The choice depends on the system and purpose of analysis. For BBH mergers, we generally use `IMRPhenomX`. For our purpose, we are using `IMRPhenomNSBH` (for asymmetric binaries where $q < 0.6$) and `IMRPhenomPv2_NRTidalv2` (for symmetric binaries⁵) to model the tidal effects in NSBH systems.⁶

The [synthetic strain data is created using power spectral densities](#) for each interferometer (H1, L1, V1).

3.2.3 Priors

Next we will define the priors uniformly for key parameters:

⁵The symmetric mixed binary mergers more or less behave as BNS mergers. So we will have to take the masses of both the components accordingly, and hence, we can't take the tidal deformability of "Black Hole" zero while using the `IMRPhenomPv2_NRTidalv2`

⁶There are so many waveform approximants available.

- **TaylorF2:** It is simple and fast, but not efficient near merger.
- **SEOBNRv4_ROM_NRTidal:** This is very accurate, but not a computationally efficient method.

For constraining NS EoS from NSBH mergers, the selected approximants are the best ones considering the computational cost, accuracy and physical realism simultaneously.

- Chirp mass ($\mathcal{M}$):

$$\text{Effective mass of the system, } \mu = \frac{m_1 m_2}{m_1 + m_2} = \frac{m_1 m_2}{M} \quad (3.9)$$

$$\text{Symmetric Mass Ratio, } \eta = \frac{\mu}{M} \quad (3.10)$$

$$\boxed{\text{Chirp Mass, } \mathcal{M} = M \eta^{3/5} = (m_1 + m_2)^{-1/5} \times (m_1 m_2)^{3/5}} \quad (3.11)$$

So, we can take a different range of chirp mass, such as, ± 0.2 , ± 0.04 etc., around the injected values to recover them in a desired range.

- Mass ratio (q):

$$q = \frac{\text{Mass of NS}}{\text{Mass of BH}} = \frac{m_1}{m_2} < 1 \quad (3.12)$$

We have taken the transition from symmetric to asymmetric system at $q = 0.6$. So, for an example, $m_1 = 4.2M_\odot$ and $m_2 = 2.8M_\odot$, $q \approx 0.66$. We can take prior as $(0.50, 0.75)^7$.

- Tidal Deformability (Λ): Define the prior range considering the computational cost. For example, for the injected value 100, a better prior range can be $(0, 200)$ instead of $(0, 500)$.
- Spins (χ): Allow aligned spins with reflective boundaries with the consideration of prograde and retrograde injected value of BH.

3.2.4 Likelihood Construction

Now, the likelihood is evaluated using `GravitationalWaveTransient` function which evaluates the fit of injected signal for a given synthetic data. For better convergence and computational cost efficiency, we marginalize it over time, phase and distance.

3.2.5 Sampling

At last, the whole parameter space is explored using either `dynesty` or `emcee` sampler to compute the posterior distributions and Bayesian evidences. For more information, see the section-3.1.3

3.3 Evidence Calculations

In this section, we will calculate the method for finding evidence for different EOS. We can compare evidences of all selected model EOS with each other and will calculate Bayes Factor (BF).

Posterior density function for NSBH mergers can be written using Bayes Theorem as:

⁷In gravitational wave signals, the phase evolution during the inspiral is primarily sensitive to the chirp mass $\mathcal{M}$, making it the best-constrained parameter. Direct sampling in component masses (m_1, m_2) introduces strong correlations, while sampling in $(\mathcal{M}, q)$ decorrelates the parameters and improves sampling efficiency.

$$p(\Theta|d_n, H_{GR}, I) = \frac{\mathcal{L}(d_n|\Theta, H_{GR}, I)p(\Theta|H_{GR}, I)}{\mathcal{Z}_{GR}} \quad (3.13)$$

here the prior on neutron star mass and Λ are between the minimum and maximum values respective to injected EOS hypothesis. Here the $\mathcal{Z}_{GR}$ is the bayesian evidence of the assumptions and can be calculated as:

$$\boxed{Z_{GR} = \int \mathcal{L}(d_n|\Theta, H_{GR}, I)p(\Theta|H_{GR}, I)d^n\Theta} \quad (3.14)$$

Following the discussion from equation-6⁸ of [Kashyap et al.](#), we can calculate the evidence in different subspaces. For the current analysis, we have used two subspaces:

1. $\tilde{\Lambda} - \eta$
2. $m_2 - \Lambda_2$

3.3.1 Evidence calculation in $(m_2 - \Lambda_2)$ space

For a hypothesized EOS k (H_k), the evidence for the $(m_2 - \Lambda_2)$ space

$$\boxed{\mathcal{Z}_{k,2} = \int dm_2 \int d\Lambda_2 \delta[\Lambda_2 - \Lambda_{k,2}(m_2)] p(m_2, \Lambda_2|d_n)} \quad (3.16)$$

The posterior of the secondary component is marginalized over the mass and its tidal deformability to get corresponding evidence in the subsequent subspace.

3.3.2 Evidence calculation in $(\tilde{\Lambda} - \eta)$ space

Instead of taking secondary parameters, we can similarly calculate evidence in the subspace of effective tidal deformability and symmetric mass ratio as follows:

$$\boxed{\mathcal{Z}_k = \int d\eta \int d\tilde{\Lambda} \delta[\tilde{\Lambda} - \tilde{\Lambda}_k(\mathcal{M}, \eta)] p(\mathcal{M}, \eta, \tilde{\Lambda}|d_n)} \quad (3.17)$$

Other subspaces can also be used based on the need of expected results, for example: $(\eta - q - \tilde{\Lambda})$, and $(m_2 - q - \Lambda_2)$ etc.

For the model selection calculation, we can find the odd ratio given as:

$$[O_j^i]_n = \frac{\mathcal{Z}(H_i|d_n, H_{GR}, I)}{\mathcal{Z}(H_j|d_n, H_{GR}, I)} \quad (3.18)$$

where H_i and H_j correspond to the hypotheses of different EOS.

Considering equal prior distributions for all EOS, the combined odd ratio can be written as:

⁸Evidence for a given equation of state k:

$$Z_k = \tilde{Z}_k \int_{U(\bar{\theta})} d^m \bar{\theta} \delta(\Theta - \theta_k(m)) p(\Theta|d_n, H_{GR}, I) \frac{p(\bar{\theta}|H_k, H_{GR}, I)}{p(\Theta|H_{GR}, I)} \quad (3.15)$$

$$\log O_j^i = \log \sum_{n=1}^N [O_j^i]_n = \sum_{n=1}^N [\log \mathcal{Z}(H_i|d_n, I) - \log \mathcal{Z}(H_j|d_n, I)] \quad (3.19)$$

but as the evidence depends on likelihood and prior (as given by eq-??, we will have to consider some correction factor.

3.4 Cost effective Computation

One of the primary thing that we should taken into account is the efficiency of the computation. From the discussion in the section-3.2, we can guess that the time taken by the **waveform generation**, **likelihood construction** and **sampling** processes can be the reason of less efficient computation.

Let's take an example to see the section-wise duration. Below are the injected and prior values:

m_1	m_2	Prior in $\mathcal{M}$	Prior in q	Λ_1	Λ_2	Prior in Λ_2
5.0	1.4	$(\mathcal{M} - 0.1, \mathcal{M} + 0.1)$	$(q - 0.1, q + 0.1)$	0	250	(200, 300)

For `dynesty` sampler with `npoints` = 700, the durations are as follows:

Processes	Duration [s]
Injection	8.8215×10^{-6}
Waveform Approximation	1.049×10^{-5}
Waveform Generation	2.451
Prior Generation	1.012
Likelihood Generation	3.595×10^{-4}
Sampling	$1.9649 \times 10^{+4}$

Table 3.2: Section-wise distribution of durations

As can be seen here, approximately, all the time is taken by the sampler. We can increase this efficiency by effectively choosing the `npoints` value and samplers.

Another study by **Hu and Veitch** [7](#) examines the computational challenges faced by standard Bayesian Parameter Estimation and its comparison with some accelerated methods. With 3G detectors in future, the detection rate and SNR will be higher, the standard methods will demand upto a **quadrillion CPU hours per month** of time. The accelerated methods like relative binning (RB), multibanding (MB), relative order quadrature (ROQ) will make the PE feasible and reduce the PE by upto **million CPU hours per month**. These methods make the estimation better, but are still very far from the ideal, as this amount of CPU hours will be a substantial burden on the computing infrastructure, electricity cost and environment.

MB method speeds up PE by adapting the frequency resolution of the GW data dynamically. As GW signals have low-frequency inspiral phases that evolve slowly, and high-frequency merger phases that evolve rapidly, MB divides the data into multiple bands with different time-frequency resolutions:

1. Low frequencies $\rightarrow$ Can be sampled at a coarser resolution (lower data rate).

2. High frequencies $\rightarrow$ Require higher resolution (finer data rate).

This significantly reduces the number of data points required without compromising accuracy. The study demonstrated that the **Multibanding (MB)** method is applicable specifically to **IMRPhenom** waveform approximants, which aligns well with our analysis framework. Therefore, we employed the MB method in our study and achieved reliable results with substantially reduced computational time.

Now, with good understanding of the bilby pipeline, bayesian inference and evidence calculations, let's look into the results.

Chapter 4

Results

Tidal deformability from the GW signals can only be detected if the **Signal-to-Noise ratio (SNR)** is very high (not less than 15-20), as the contribution of $\tilde{\Lambda}$ in phase evolution is introduced only after the 5th order of Post Newtonian Approximation (5PN). So, to constrain the value of $\tilde{\Lambda}$ (or Λ), we need the SNR to be high.

In the initial runs, we injected random parameters for both the 4CBC and 15CBC datasets, which led to signal-to-noise ratios (SNRs) falling outside the desired range. In subsequent runs, we modified the injection settings to account for and improve the SNR distribution.

As we discussed earlier, we can use **IMRPhenomPv2_NRTidalv2** as the waveform approximant for symmetric binaries, as the system can mimic the effects of Binary Neutron Star merger. Also, for asymmetric system, we can use **IMRPhenomNSBH** approximant. The important thing to note here is that for an asymmetric system, we can fix the tidal deformability of one of the components (which is the black hole) to be zero. On the other hand, in the case of symmetric masses, we can't fix the Λ 's, instead, we will have to make them free and independent of each other.

4.1 4 CBC runs

4.1.1 Manual runs

So, let's first start with the 4cbc runs. Initially, we did some runs with manual injection parameters for both the approximants. Below is the summary of the following runs:

	Approximant	m_1	m_2	$\mathcal{M}$	$\mathbf{q}$	Λ_2	$\tilde{\Lambda}$	Network SNR
Fig-4.1,4.2	IMRPhenomNSBH	5.5	1.4	2.31	0.25	190	3.87	42.23
Fig-4.3,4.4	IMRPhenomPv2_NRTidal	5.0	3.0	3.35	0.6	40	7.67	56.69

Table 4.1

Here, in Figure-4.1, except Λ_2 , every parameter is tightly constrained and recovered very well. The primary reason may be the insufficient number of sample points, which resulted in poor coverage of the parameter space. Another contributing factor to the poor recovery of Λ could be the prior range, which may not sufficiently encompass the true parameter values.

Comparing figures 4.2a and 4.2b, we observe that transforming from (m_1, m_2, Λ_2) to $(\mathcal{M}, q, \tilde{\Lambda})$ results in a smoother and tighter posteriors, especially in $\tilde{\Lambda}$. We can also see a

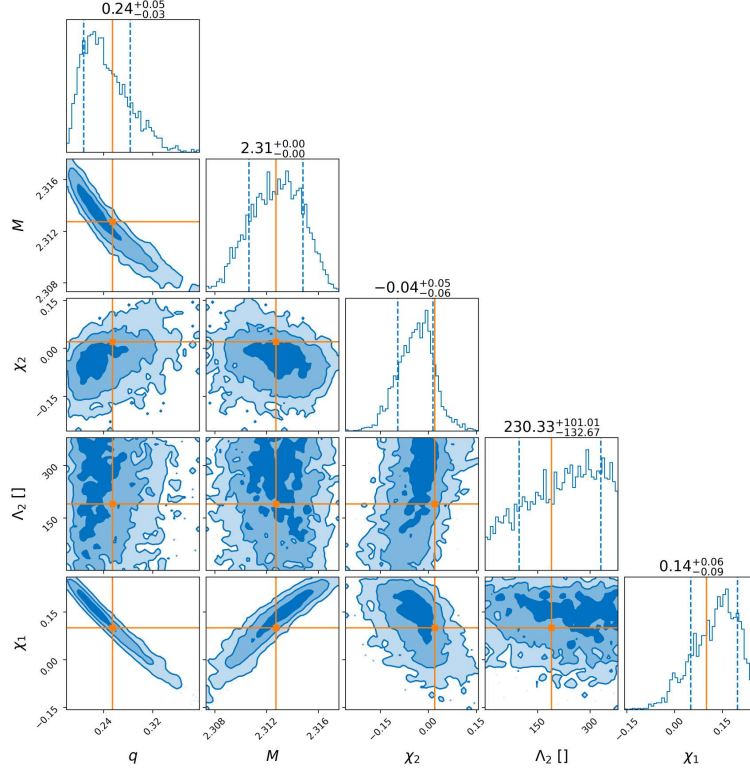


Figure 4.1: Posterior distribution using IMRPhenomNSBH.

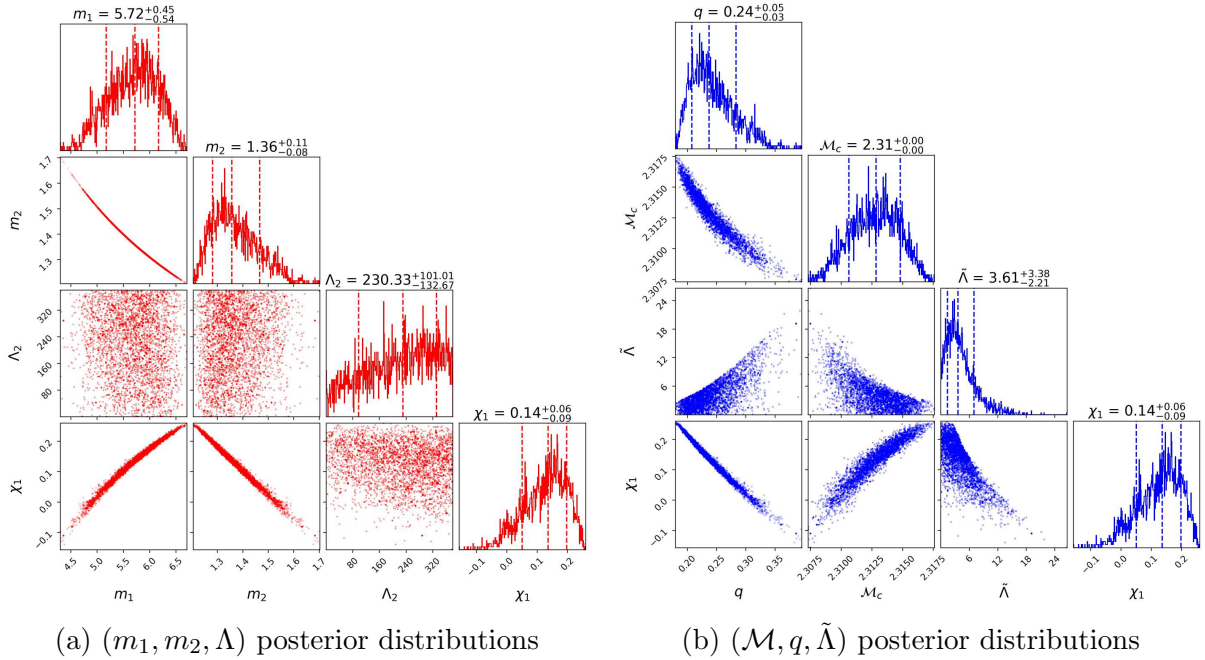


Figure 4.2

clearer representation of physical correlations between $\mathcal{M}$ and $\tilde{\Lambda}$. This reflects how waveform models encode tidal effects predominantly through the effective tidal deformability, which is a mass-weighted average and hence, more directly constrained.

As with the IMRPhenomNSBH approximant, Figures 4.3 and 4.4 present the posterior distributions obtained using IMRPhenomPv2.NRTidal, based on the injected parameters

provided in Table 4.1.

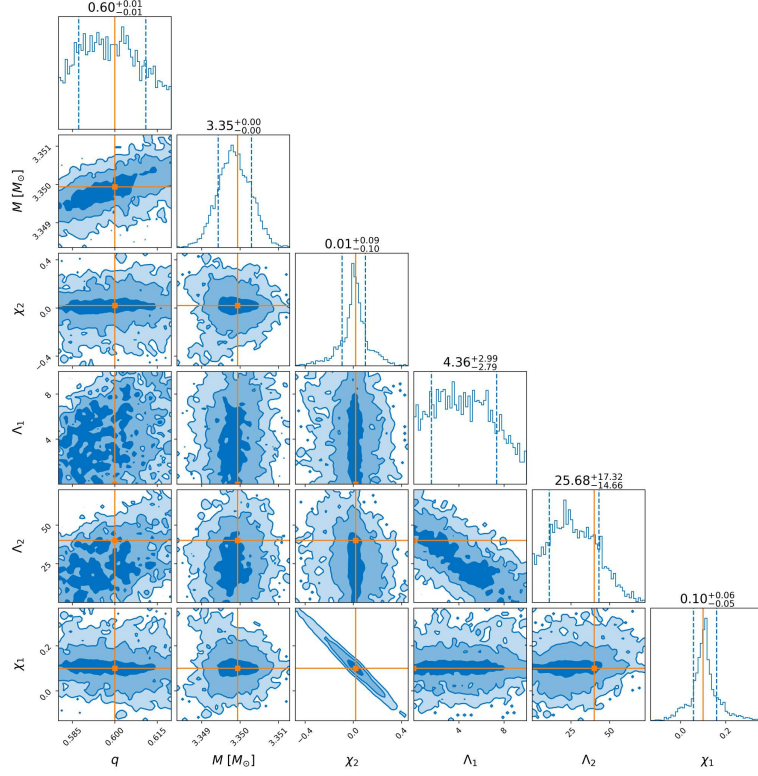


Figure 4.3: Posterior distribution using IMRPhenomPv2_NRTidal.

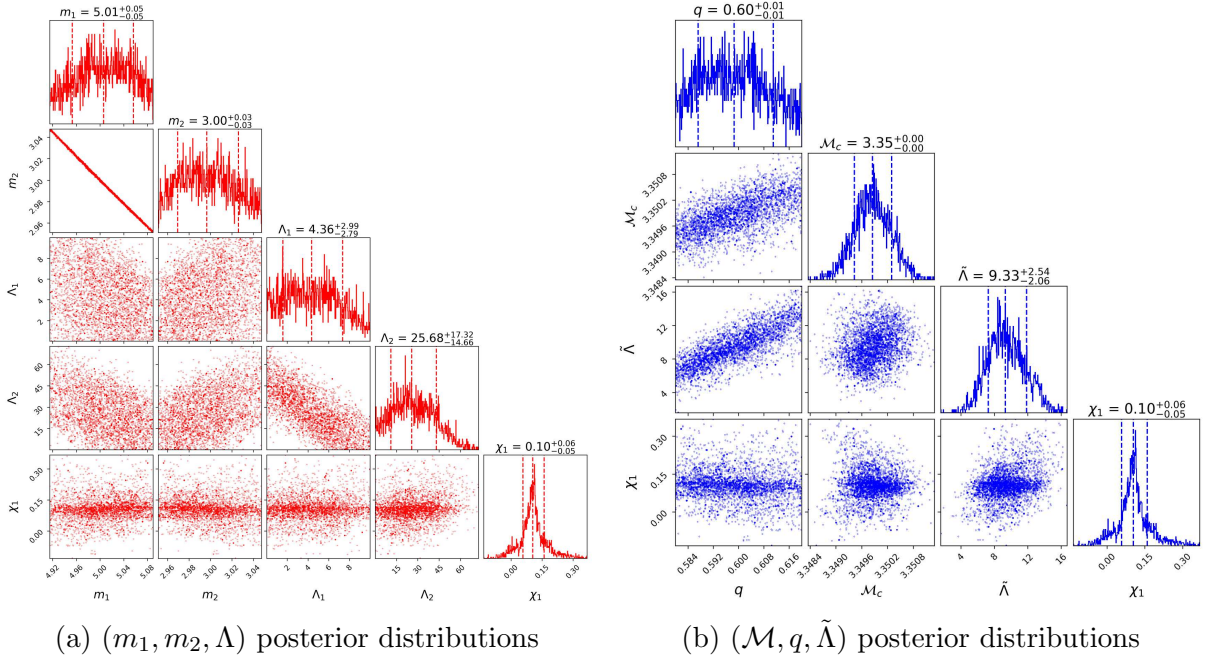


Figure 4.4

Other than minor uncertainties, the overall posterior distribution shows slightly narrower contours than in the IMRPhenomNSBH case, particularly in the $\mathcal{M}$ - Λ_2 plane. This

indicates a better correlation capture due to more physically complete waveform modeling.

4.1.2 Automation

After performing a sufficient number of manual injection runs, we developed an automation script that selects random binary component masses (within an optimal range) and calculates the corresponding tidal deformability of the neutron star for the injected mass, using different equations of state (APR4, SLy, DD2).

Using various cost-effective computational techniques, we generated simulated data for three different equations of state: **APR4** (30 runs), **SLy** (10 runs), and **DD2** (10 runs). As mentioned above, the component masses are generated randomly but uniformly within the chosen range.

Additionally, we have set a threshold mass ratio of $q = 0.6$, where different waveform approximants can be applied. However, the probability of obtaining binary systems with a mass ratio above 0.6 is significantly lower than that of obtaining systems with $q < 0.6$, i.e.;

$$\boxed{P(q > 0.6) \ll P(q < 0.6)} \quad (4.1)$$

Figure 4.5 shows the recovered tidal deformability Λ_2 and corresponding radius R_2 for a range of neutron star masses m_2 , across three equations of state. The expected inverse relationship between Λ_2 and m_2 is clearly observed, consistent with the physical notion that [heavier neutron stars are more compact and thus less tidally deformable](#).

However, the recovered values of Λ_2 exhibit large and asymmetric error bars. This is likely due to the relatively low signal-to-noise ratios (SNRs) in some of the injected runs, which degrade the sensitivity to tidal effects.

The lower panel shows the corresponding neutron star radius R_2 derived from each EoS. These trends remain consistent with theoretical expectations—DD2 (stiff) produces larger radii, while APR4 (soft) yields smaller ones. The results validate the transformation from Λ_2 to R_2 using EoS-dependent mass-radius relations.

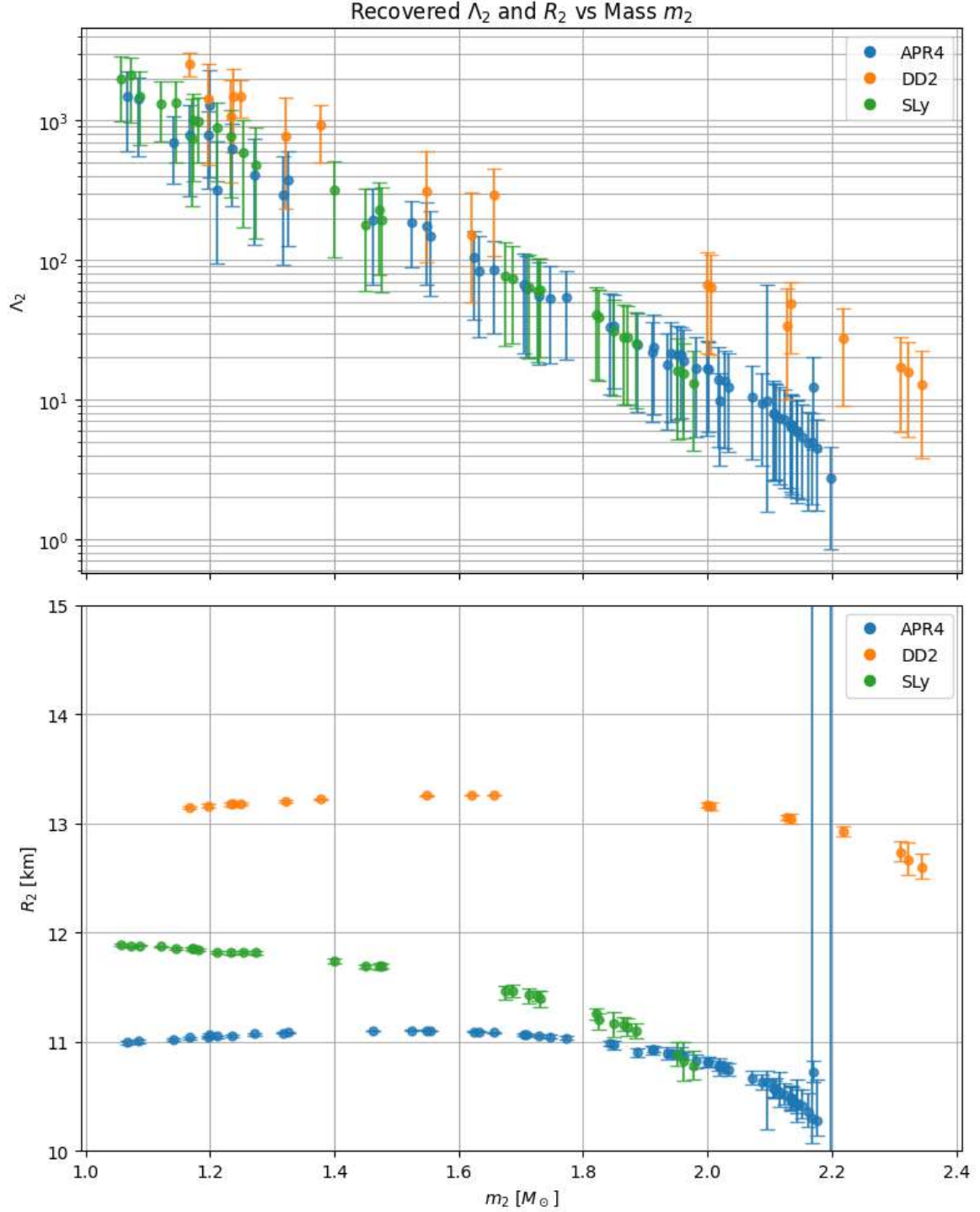


Figure 4.5: Constraining 3 different EoS for 4-parameter runs.

4.2 15 CBC runs

After doing parameter estimation for minimal runs (4 parameters), we simulated data over all parameter space. The simulations are done for 3 injected EoS (APR4, SLy and DD2) with 3 different detector configurations (O5, $A\sharp$ and ECC).

In Figure 4.6, we present the relative error between the injected chirp mass values and the corresponding posterior estimates for different equations of state (EoS).

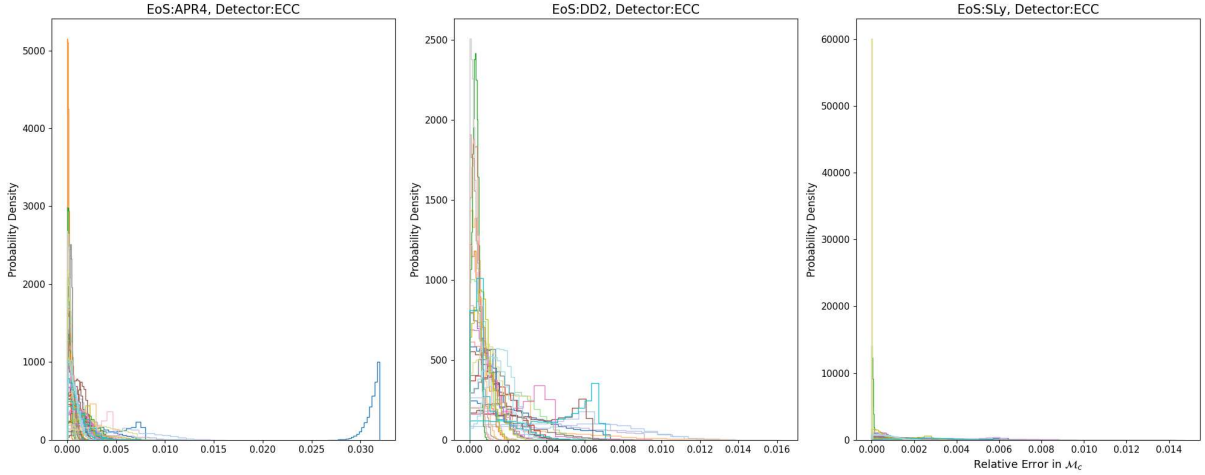


Figure 4.6: Relative Error in Chirp Mass of simulated binaries.

Now, we compared the injected value of Λ_2 with its generated posterior distributions. Although they are not good results. The main reason is the low values of SNR for different runs.

Similar to the figure-4.6, we did the efficiency check for tidal deformability as shown in figure-4.7. This is shown for ECC detector. The expected distribution is the truncated gaussian with peak at 0.

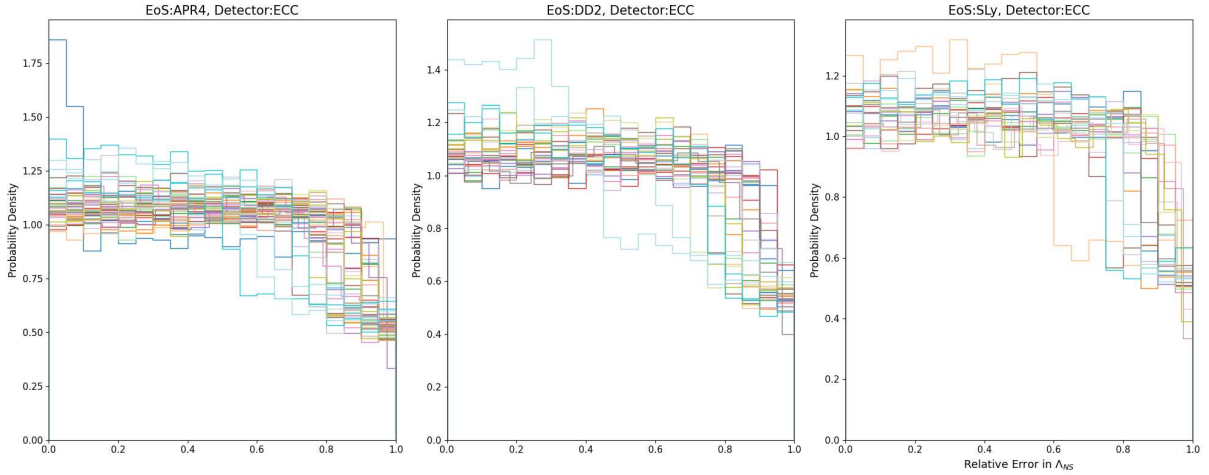


Figure 4.7: Relative Error in Tidal Deformability of neutron star in the simulated binaries.

The primary deviation from the expected distribution can be attributed to low signal-to-noise ratios (SNRs). Even after applying an SNR threshold, insufficient sampling may also contribute to the deviation by failing to adequately explore the parameter space.

Figure 4.8 presents the recovered tidal deformability parameter Λ_2 as a function of the neutron star mass m_2 for several injected NSBH systems using three different equations of state (APR4, SLy, DD2). The general decreasing trend of Λ_2 with increasing m_2 is consistent with theoretical predictions. However, significant scatter and large uncertainties are observed reflecting the fact its contribution to the waveform enters at high PN order, making it difficult to recover in low-SNR or weakly tidal systems.. This behavior is attributed to the limited sensitivity of gravitational waveforms to tidal effects in resolving the sharp cutoff near the EoS-dependent maximum neutron star mass.

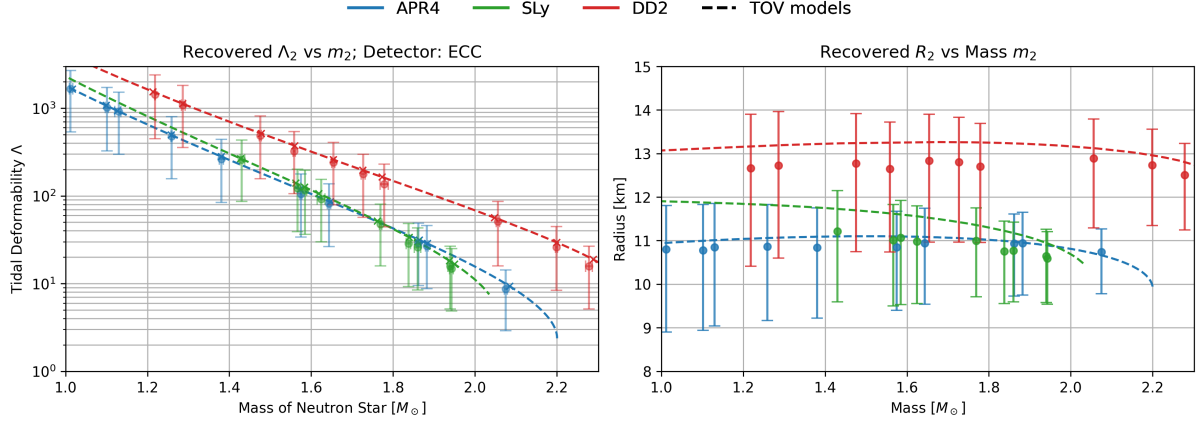


Figure 4.8: Constraining 3 different EoS for 15-parameter NSBH runs for ECC detector configuration.

Figure-4.9 shows distributions of some of the key parameters with their correlation with others with **APR4** as the injected EoS. Here the $\bullet$ are the injected values and their probability distributions are shown in black-dashed histogram in diagonal boxes. The red, blue and green contours and probability distributions are showing the recovered posteriors from **ECC**, **A[‡]** and **O5** detectors. The table in the upper-right corner shows the median and 1- σ values of injected and recovered values for the given parameters.

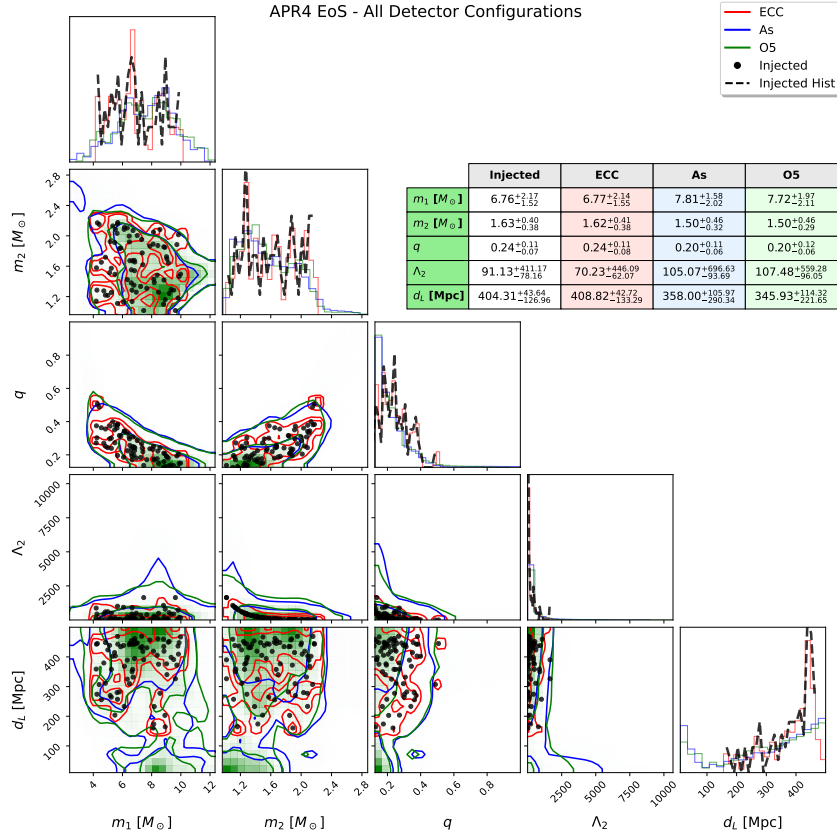


Figure 4.9: Injected (in dashed lines) and recovered distributions (in solid lines) of component masses, tidal deformability of neutron star, mass ratio and the luminosity distance for the case of APR4 as the injected EoS and all three detectors.

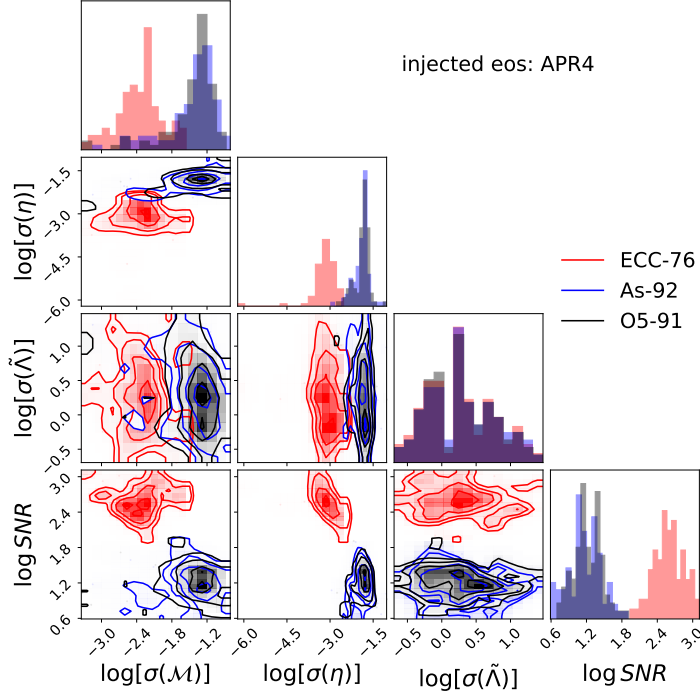
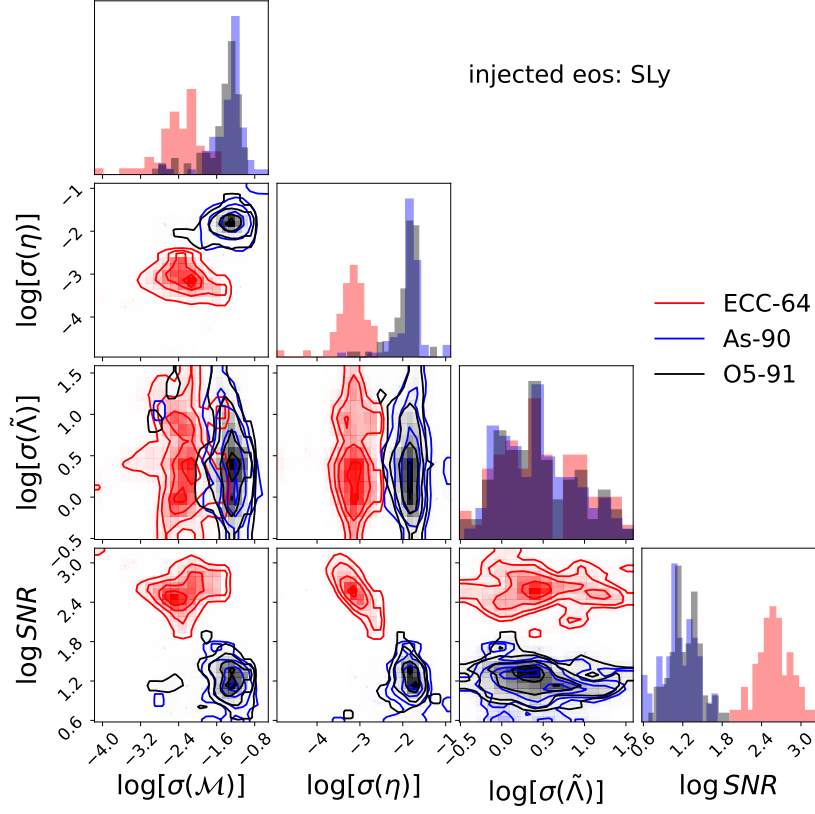


Figure 4.10: The error distributions of key parameters of binary mergers: $\mathcal{M}$, η and $\tilde{\Lambda}$. The error in chirp mass and symmetric mass ratio for the O5 and $A^\sharp$ detectors are almost similar, which is further improved by the next generation ECC detector by almost an order of magnitude. The tidal effects are introduced in the 5th PN order, hence they are very difficult to recover. But it was expected from ECC to show a similar kind of error distribution as of $\mathcal{M}$ and η , even with a lesser number of events, but all three detector configurations showed almost similar trend for tidal deformability.

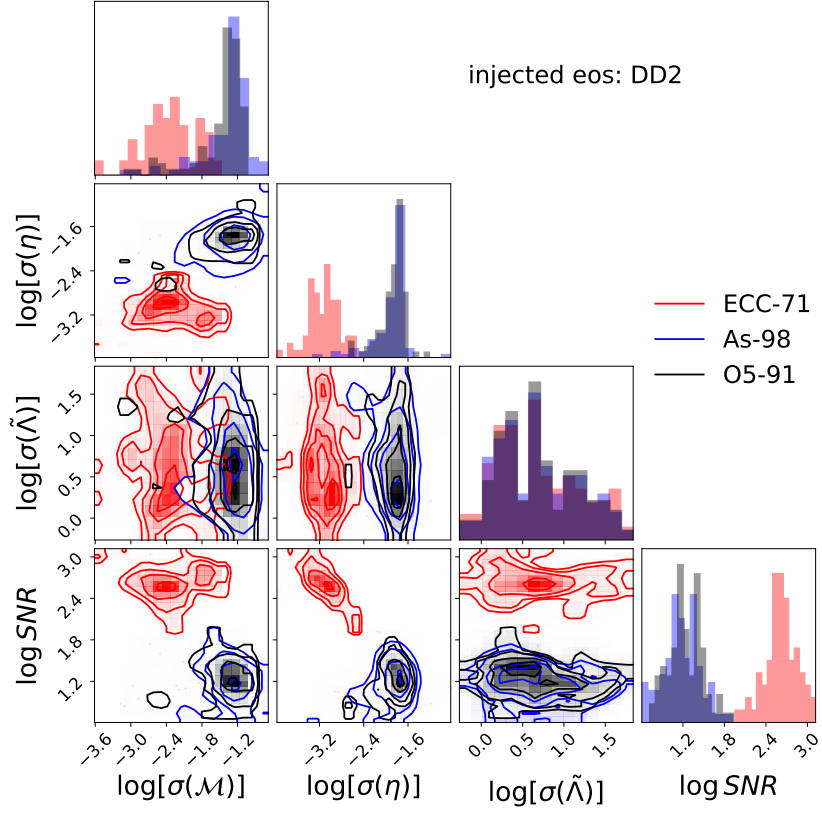
In figure-4.10, we have shown the error distribution of 3 fundamental parameters of neutron star: $\mathcal{M}$, η and $\tilde{\Lambda}$. It can be seen that $O5$ and $A^\sharp$ are showing similar kind of distribution in errors, while the *ECC* has significant improvement in measuring at least first 2 of the 3 parameters (The numbers in the legend are number of events corresponding to that detector configuration. As *ECC* has lesser number of events compared to other two, may be that can be the reason of non-distinguishable distribution of $\sigma(\tilde{\Lambda})$). The *ECC* detector also increased the sensitivity by 2 to 3 orders of magnitude in comparison to $O5$ and $A^\sharp$ detectors.

Similar error distributions are shown by other 2 injected EoS. This can be seen in figure-4.11.

We calculated the evidence in two different parameter subspaces: $(\eta - \tilde{\Lambda})$ and $(m_2 - \Lambda_2)$ as explained in 3.3.1 and 3.3.2. The corresponding evidence distributions are shown in fig-4.2. Here, the injected eos is APR4 for all 3 detector configurations. The first two panels correspond to $O5$ and $A^\sharp$, respectively. In both of the parameter subspaces, the distribution is nearly the same for both configurations. This trend is changed with *ECC*, as the $\tilde{\Lambda} - \eta$ subspace still shows similar distributions, but all the distributions in $m_2 - \Lambda_2$ have vanished.



(a)



(b)

Figure 4.11: Error distributions of $\mathcal{M}$, η , $\tilde{\Lambda}$ and network SNR for SLy and DD2 as injected EoS.

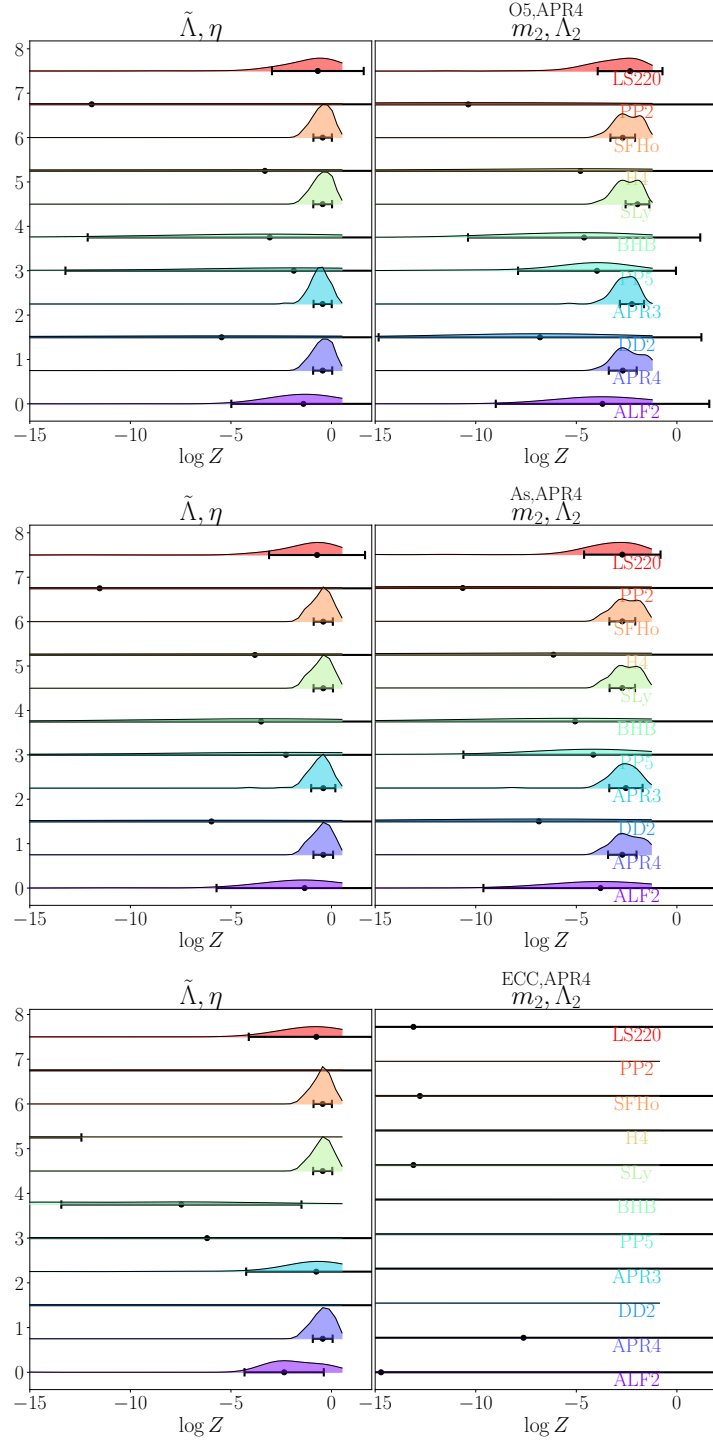


Figure 4.12: In these figures, we have shown the evidence distribution using the two methods discussed in section-3.3 with APR4 as injected EoS. The three panels are showing distributions for all three detector configurations: (a) $O5$, (b) $A^\sharp$, (c) ECC ; with median and $1 - \sigma$ error values. It can be seen from first two figures they didn't show much difference to distinguish EoS. Although, ECC should've shown an improvement, it is also not showing any. Important point to notice is that for $m_2 - \Lambda_2$ subspace, $APR4$ has the highest $\log Z$. Similar trend is shown for other EoS also.

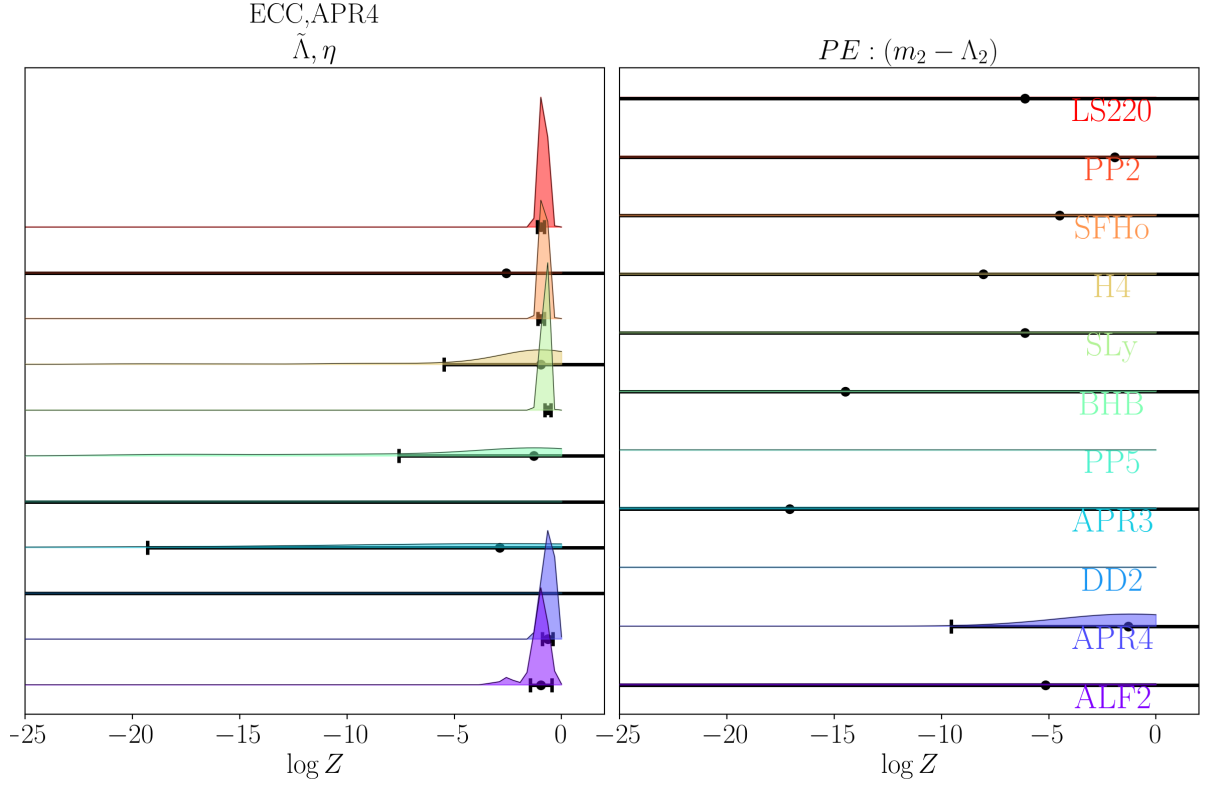


Figure 4.13: Evidence Distribution of 30 random nearly symmetric simulated events ($q \geq 0.6$). The distribution is shown for APR4 as injected EoS and ECC as the detector configuration.

Although, nearly symmetric systems show the expected distribution. For the first parameter subspace ($\tilde{\Lambda} - \eta$), there are at least 5 model EoS (LS220, SFHo, SLy, APR4 and ALF2) which are showing the non-zero recovery of the injected EoS, and nearly all of the model EoS has equal value of median of evidence. In the second subspace ($m_2 - \Lambda_2$), the differences are clearly visible. Only the APR4 has a non-zero evidence distribution with maximum median value (PP2 has also same value as of APR4, but PP2 has null distribution). So, as expected, the injected EoS is recovered very well in the ($m_2 - \Lambda_2$) subspace.

These inconsistencies highlight the limitations of current parameter estimation. They emphasize the need for improved waveform models, enhanced SNR conditions, and better control over injected population parameters for more reliable recovery of Λ_2 .

Chapter 5

Conclusions

In this project, we carried out a systematic simulation-based analysis aimed at constraining the neutron star equation of state (EoS) using gravitational wave signals from NSBH mergers. A major portion of the work was dedicated to **optimizing the inference pipeline** by carefully **tuning the MCMC sampler parameters** through repeated trial-and-error runs. This included setting appropriate priors, adjusting the number of live points, and refining the sampler configuration in order to achieve efficient convergence without unnecessary computational overhead.

To make the analysis computationally cost-effective, we employed several key techniques, including frequency-domain waveform selection, **multibanding**, and parallelized simulation scripting. Although full implementation of **relative binning** and **reduced order quadrature (ROQ)** was beyond the scope of the current phase, these remain promising extensions for future work.

We performed simulations over two ensembles of injection sets — one with **4 CBC runs for baseline testing**, and a second with **15 CBC runs varying mass ratios, spin configurations, and inclination angles**. The recovered Λ_2 versus m_2 relation showed the expected physical trend: **tidal deformability decreases with increasing neutron star mass**. However, several data points exhibited large uncertainties, primarily due to the maximum mass limitations imposed by certain EoS models, and more significantly, **due to low SNR in the injected signals**.

For 3 different EoS (APR4, SLy and DD2) and with 3 future detector configurations (O5, $A\sharp$ and ECC), we calculated the error distributions of the key parameters. ECC showed significant improvement in the recovery of those parameters, except tidal deformability (the expectation was to show at least an order of improvement compared to O5 and $A\sharp$). The evidence calculation and evidence distribution in different subspaces is used to find the maximum value for model eos. Our analysis showed that for symmetric binaries, the $(m_2 - \Lambda_2)$ subspace showed the only non-zero distribution of evidence when injected and model eos are same, while it showed null distribution for all other eos. There are some aspects and future study needed to be done, which we have discussed in Chapter-6

To visualize the overall constraints on tidal deformability, we compared posterior distributions of Λ_2 with the injected values, and **found noticeable inconsistencies across runs**. These inconsistencies are attributed almost entirely to insufficient signal-to-noise ratio, which severely limits the ability to extract tidal signatures encoded in the late inspiral phase of the waveform. Despite this, the current study has allowed us to **identify favorable parameter regimes and waveform configurations** that can enable more reliable inference in future work.

Chapter 6

Future Work

Based on the insights gained from this study, several directions have emerged for future work that aim to improve both the accuracy of parameter recovery and the computational efficiency of the NSBH inference pipeline.

A major priority is to incorporate the `SEOBNRv2` waveform approximant, which provides a more accurate modeling of the inspiral and tidal interactions, particularly in the strong-field regime. `SEOBNR` models, derived from the Effective One Body formalism, are known to perform well in parameter recovery—especially for asymmetric binaries and configurations involving moderate spin. Using these in tandem with `IMRPhenomPv2.NRTidal` will allow us to cross-validate results and potentially reduce systematic biases arising from waveform systematics.

Additionally, the low SNR limitation that hampered the recovery of tidal deformability in this study will be addressed. We will simulate higher SNR injections or injecting into realistic noise at varying distances. Also, we will stack multiple low-SNR events to achieve population-level constraints,

Another key goal is to refine the parameter space sampling by incorporating astrophysically informed priors and testing the impact of realistic neutron star spins, misalignment, and possibly eccentricity. We also plan to extend the simulation study to larger sets of NSBH binaries and explore population-level constraints on the EoS.

The main focus for our future goal is to extend the study of evidence calculations in few more parameter subspaces, as the distributions can be different for different subspaces. In the case of [Kashyap et al.](#) study, they took 3 spaces ($\mathcal{Z}(\eta - \tilde{\Lambda})$, $\mathcal{Z}(m_1 - \Lambda_1 - m_2 - \Lambda_2)$, $\mathcal{Z}(m_1 - \Lambda_1) + \mathcal{Z}(m_2 - \Lambda_2)$). We will also try some new spaces, for example $(\eta - q - \tilde{\Lambda})$ or $(m_2 - q - \Lambda_2)$. Maybe in these spaces, the general population of NSBH will also show the expected behaviour of evidence distributions.

In addition to that, we are currently trying to get the cumulative evidence and Bayes' Factor of different model eos for model selection. That will lead to the quantitative conclusion of the constraining equation of state. It can also tell us that for next generation detectors, how many NSBH events are required to totally conclude the equation of neutron stars and their tidal properties.

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